



UK Health
Security
Agency

Epidemiology Modelling Review Group: consensus statement on COVID-19

Date: 27 April 2022

Introduction

The UK Health Security Agency (UKHSA) Epidemiology Modelling Review Group (EMRG) shares this consensus statement on COVID-19 with acknowledgment to SPI-M-O, who have developed and shared modelling methodologies and contribute model outputs to these combined estimates.

All probability statements are in line with the framework given in [Annexe A](#).

Summary

1. UKHSA's best estimate for R in England is between 0.7 and 0.9. R is estimated to be between 0.7 and 1.0 for Wales, 0.8 and 1.0 for Scotland, and 0.7 and 0.9 for Northern Ireland ([Figure 1](#)). These estimates are based on models¹ fit to data available up to 25 April 2022 where feasible, including hospitalisations, deaths, testing, wastewater samples and longitudinal studies.
2. Changes to testing policy and guidelines (including suspensions of free PCR and LFD testing for the general public) were made in England² and Wales³ on 1 April 2022. Changes to testing policy and guidelines are being made in Northern Ireland⁴ and Scotland⁵. Changes to testing guidance alter testing data streams, and model combinations used to provide R and growth rate estimates have been adjusted where required, to reduce the use of these data.
3. Estimates lag changes in transmission by 2 to 3 weeks, due to the time required to see changes in data streams. It is an average over time, geographies, viral variants, and communities.
4. Combined estimates⁶ show that the incidence⁷ is between 137,000 and 223,000 new infections per day in England.

¹ Model estimates are required as quantities such as the Reproduction Number (R) are not directly observable. Instead, a variety of independently produced models are used to interpret the data and estimate R. The combination of models able to be included can change between weeks and therefore care should be taken when drawing week-on-week comparisons.

² [Changes to COVID-19 testing in England from 1 April - GOV.UK \(www.gov.uk\)](#)

³ [Changes to testing and care homes as we all learn to live safely with coronavirus | GOV.WALES](#)

⁴ [Test-Trace-Transition-Plan.pdf \(health-ni.gov.uk\)](#)

⁵ [Coronavirus \(COVID-19\): Test and Protect - transition plan - gov.scot \(www.gov.scot\)](#)

⁶ Different nations and regions may use different sets of models for these estimates; hence caution should be applied in drawing direct comparisons. For example, fewer models produce estimates for Wales and Northern Ireland.

⁷ The number of new infections per day.

Incidence and prevalence

5. During its most recent week (ending 23 April 2022), the ONS COVID-19 Infection Survey estimates⁸ that an average of 2,408,300 people had COVID-19 in the community in England (95% credible interval 2,326,300 to 2,494,500). The survey does not include people in care homes, hospitals or prisons. Estimates from across the 4 nations of the UK are:
- England 2,408,300 (95% credible interval 2,326,300 to 2,494,500)
 - Scotland 218,000 (95% credible interval 193,800 to 243,600)
 - Wales 172,300 (95% credible interval 151,300 to 194,700)
 - Northern Ireland 74,700 (95% credible interval 60,500 to 91,700)

Growth rate and reproduction number

For small daily changes, the growth rate is approximately the proportion by which the number of infections increases or decreases per day, that is, the speed at which an epidemic is growing or shrinking. However, at very high growth rates, this relationship does not hold.⁹

6. The EMRG's consensus estimates for the growth rates in the 4 nations are (90% credible interval):
- England is between -6% to -3% per day
 - Wales is between -7% to -1% per day
 - Scotland is between -5% to -1% per day, and
 - Northern Ireland is between -5% to -2% per day

National estimates of growth rates are summarised in [Figure 2](#).

7. The reproduction number (R) is the average number of secondary infections produced by a single infected individual; it is an average over time, geographies, viral variants, and communities.
8. UKHSA's best estimate for R in England is 0.7 and 0.9. R is estimated to be between 0.7 and 1.0 for Wales, 0.8 and 1.0 for Scotland, and 0.7 and 0.9 for Northern Ireland. UKHSA's agreed national and regional R estimates are summarised in [Table 1](#), [Table 2](#) and individual model estimates are illustrated in [Figure 1](#), and [Figure 3](#).
9. Changes to testing policy and guidelines (including suspensions of free PCR and LFD testing for the general public) were made in England¹⁰ and Wales¹¹ on 1 April 2022. Similar

⁸ These estimates can be subject to revision as further information is available and modelled.

⁹ Further Technical Information on the growth rate can be found in Plus Magazine: [The growth rate of COVID-19 | plus.maths.org](https://plus.maths.org/).

¹⁰ [Changes to COVID-19 testing in England from 1 April - GOV.UK \(www.gov.uk\)](https://www.gov.uk/government/news/changes-to-covid-19-testing-in-england-from-1-april)

¹¹ [Changes to testing and care homes as we all learn to live safely with coronavirus | GOV.WALES](https://gov.wales/government/news/changes-to-testing-and-care-homes-as-we-all-learn-to-live-safely-with-coronavirus)

changes to testing policy and guidelines are being made in Northern Ireland¹² and Scotland¹³. Changes to testing guidance alter testing data streams, and model combinations used to provide R and growth rate estimates have been adjusted where required, to reduce the use of these data. Care should be taken when comparing with previous estimates.

10. R is an indicator that lags changes in transmission by 2 to 3 weeks,¹⁴ due to the time required for changes to be seen in data streams. Therefore, while epidemic estimates for R and other metrics such as growth rate, use the latest data available up to 25 April 2022, where feasible,¹⁵ the estimates reported here represent the epidemic situation as at 12 April 2022.
11. Estimates of R and the growth rates per day become less useful in determining the state of the epidemic when there is a high degree of immunity to the circulating variant in the population. Particular care should be taken when interpreting these estimates.
12. In addition, changes in population immunity can impact data streams, with alterations to the relationship between cases and health outcomes, such as hospital admissions. These relationships will continue to change depending on the degree of immune protection to variant(s) in circulation.
13. R and growth rate estimates indicate the magnitude of growth or decay of the epidemic. However, these indicators should be considered alongside other measures of the epidemic, such as incidence,¹⁶ and prevalence.¹⁷ When prevalence is very high, even if R is at, or below, 1, the absolute number of new cases will be very high.

¹² [Test-Trace-Transition-Plan.pdf \(health-ni.gov.uk\)](#)

¹³ [Coronavirus \(COVID-19\): Test and Protect - transition plan - gov.scot \(www.gov.scot\)](#)

¹⁴ Different data-streams and different models are expected to be lagged in their estimates by different amounts when compared with the true underlying epidemiological situation. This is due to multiple lags such as reporting and delays in the infection processes.

¹⁵ Different models fit to different windows of time using different methodologies, hence not all models will fit up to this precise date.

¹⁶ The number of individuals who develop the disease within a specified time period

¹⁷ The proportion of the population with the disease at a given point in time

Table 1. Combined estimates of R values growth rates and doubling times in the 4 nations of the UK (90% credible interval)

Nation	R	Daily growth rate	Doubling time ¹⁸
England	0.7 to 0.9	-6% to -3%	-21 to -12 days
Wales	0.7 to 1.0	-7% to -1%	-11 days to flat
Scotland	0.8 to 1.0	-5% to -1%	-15 days to flat
Northern Ireland	0.7 to 0.9	-5% to -2%	-31 to -13 days

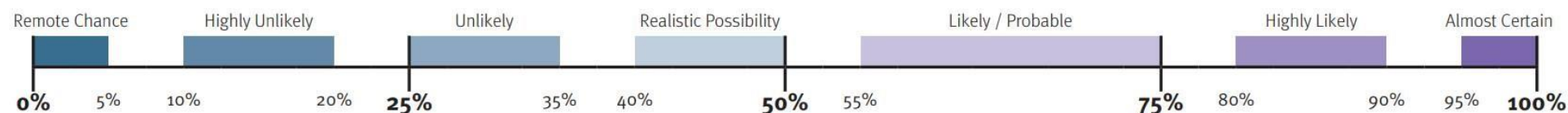
Table 2. Combined estimates of R values growth rates and doubling times in the NHS England regions (90% credible interval)

NHS England region	R	Daily growth rate	Doubling time ¹⁸
England	0.7 to 0.9	-6% to -3%	-21 to -12 days
London	0.7 to 0.9	-7% to -2%	-24 to -11 days
East of England	0.7 to 0.9	-6% to -2%	-25 to -12 days
Midlands	0.7 to 0.9	-7% to -3%	-21 to -11 days
North East & Yorkshire	0.7 to 0.9	-6% to -2%	-31 to -13 days
North West	0.6 to 0.9	-8% to -4%	-16 to -9 days
South East	0.6 to 0.9	-7% to -3%	-19 to -11 days
South West	0.7 to 0.9	-6% to -3%	-22 to -11 days

¹⁸ Any estimates with a halving or doubling time of more than 40 days have been described as flat. Negative values of doubling time indicate a halving time (the time expected for cases to fall by 50%). Doubling time here is calculated using the growth rate.

Annexe A. PHIA framework of language for discussing probabilities

The yardstick splits the probability scale into 7 ranges from remote chance (0 to 5% probability) to almost certain (95% to 100% probability).



Acknowledgements

UKHSA takes responsibility for this consensus statement and its contents. However, UKHSA would like to acknowledge the work of SPI-M-O and academic partners in developing methodologies and sharing these, as well as continuing to contribute model outputs to the combined estimates. These estimates include contribution from LSHTM ([1](#), [2](#)), Imperial College London ([3](#), [8](#)), University of Warwick ([4](#), [5](#)), University of Exeter and University of Bristol ([6](#)), Lancaster University ([7](#)), University of Manchester and University of Cambridge ([9](#)). UKHSA would also like to thank the European Bioinformatics Institute ([10](#)), University of Oxford ([11](#), [12](#)), University of Liverpool ([13](#)), and the Institute of Disease Modeling ([14](#)) for contributing model outputs. UKHSA also acknowledges the work developing combination estimates from Defence and Science Technology Laboratory ([15](#)). UKHSA also thanks and acknowledges the support and collaboration of the SPI-M-O Secretariat and co-Chairs, as well as colleagues across the 4 nations.

References

1. Abbott, Hellewell and others [‘Estimating the time-varying reproduction number of SARS-CoV-2 using national and subnational case counts’](#). Wellcome Open Research, 8 December 2020
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15. Maishman and others. [‘Statistical methods used to combine the effective reproduction number, \$R\(t\)\$, and other related measures of COVID-19 in the UK.’](#) arXiv preprint, 3 March 2021

Figure 1a. Estimates of R in the 4 UK nations (90% credible intervals) as at 12 April 2022

Bars represent different independent estimates. The grey shaded areas represent the combined numerical range and the black bars are the combined range after rounding outwards to one decimal place.

England

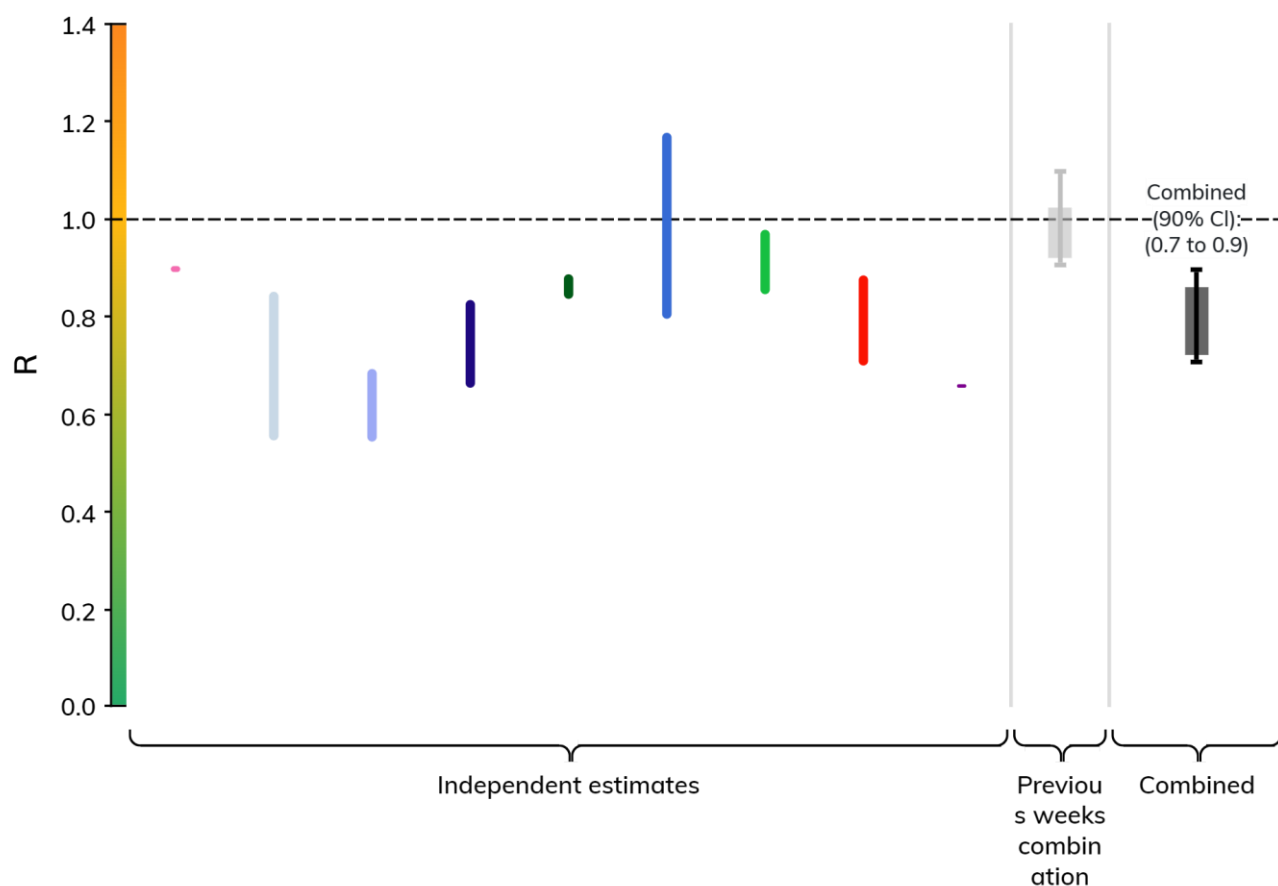


Figure 1b. Estimates of R in the 4 UK nations (90% credible intervals) as at 12 April 2022

Bars represent different independent estimates. The grey shaded areas represent the combined numerical range and the black bars are the combined range after rounding outwards to one decimal place.

Wales

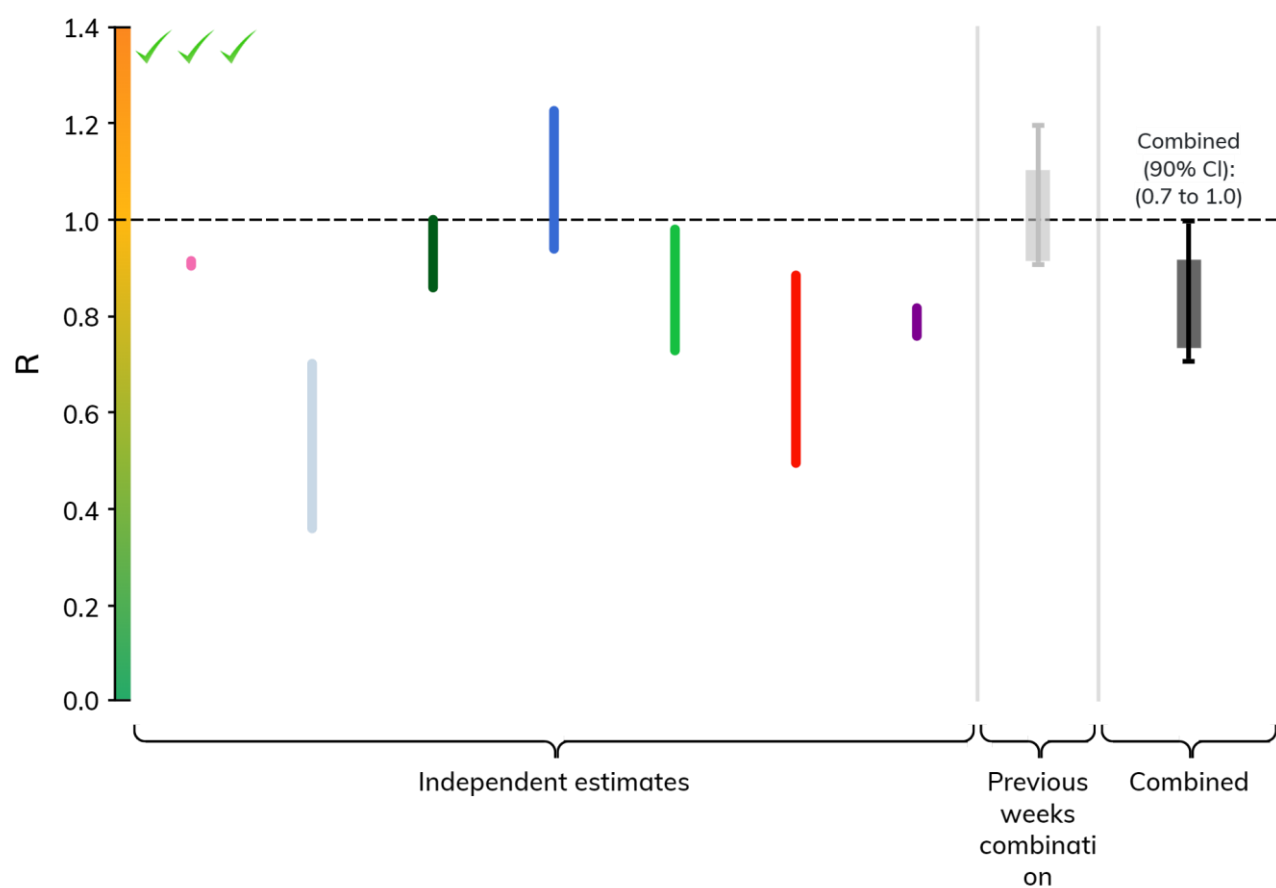


Figure 1c. Estimates of R in the 4 UK nations (90% credible intervals) as at 12 April 2022

Bars represent different independent estimates. The grey shaded areas represent the combined numerical range and the black bars are the combined range after rounding outwards to one decimal place.

Scotland

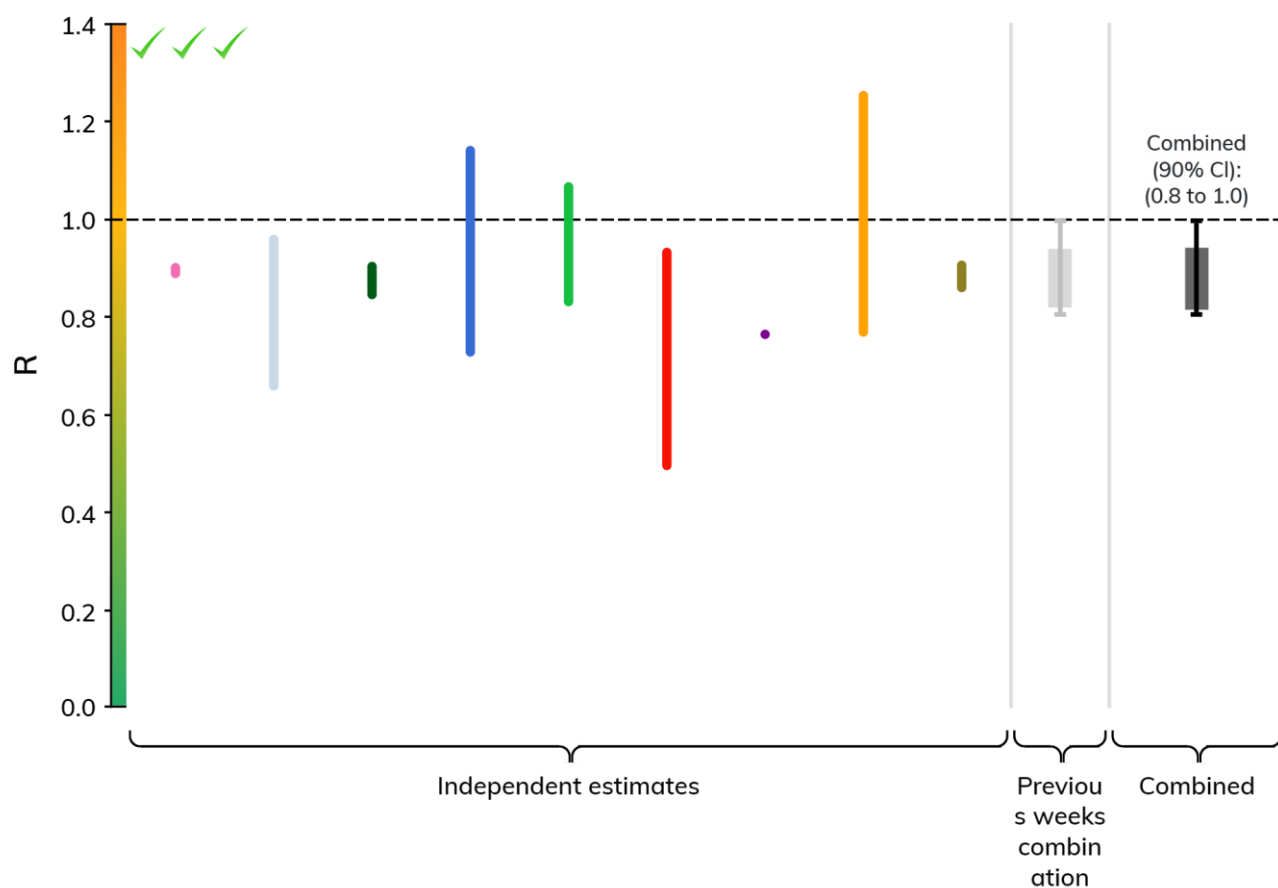
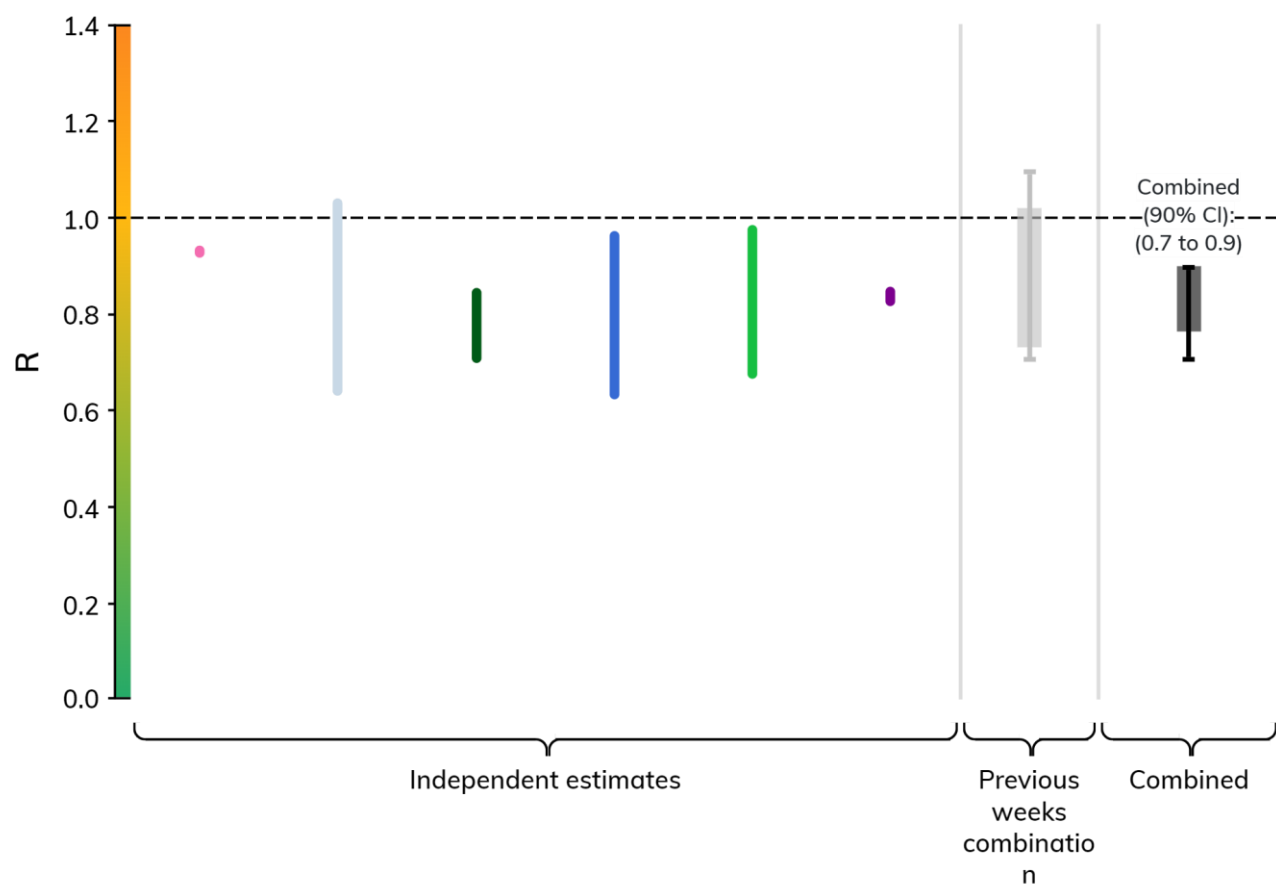


Figure 1d. Estimates of R in the 4 UK nations (90% credible intervals) as at 12 April 2022

Bars represent different independent estimates. The grey shaded areas represent the combined numerical range and the black bars are the combined range after rounding outwards to one decimal place.

Northern Ireland



England



Figure 2b. Estimates of the growth rate in the 4 UK nations, including 90% credible intervals as at 12 April 2022

Bars represent different independent estimates. The grey shaded areas represent the combined numerical range and the black bars are the combined range after rounding outwards to the nearest per cent.

Wales

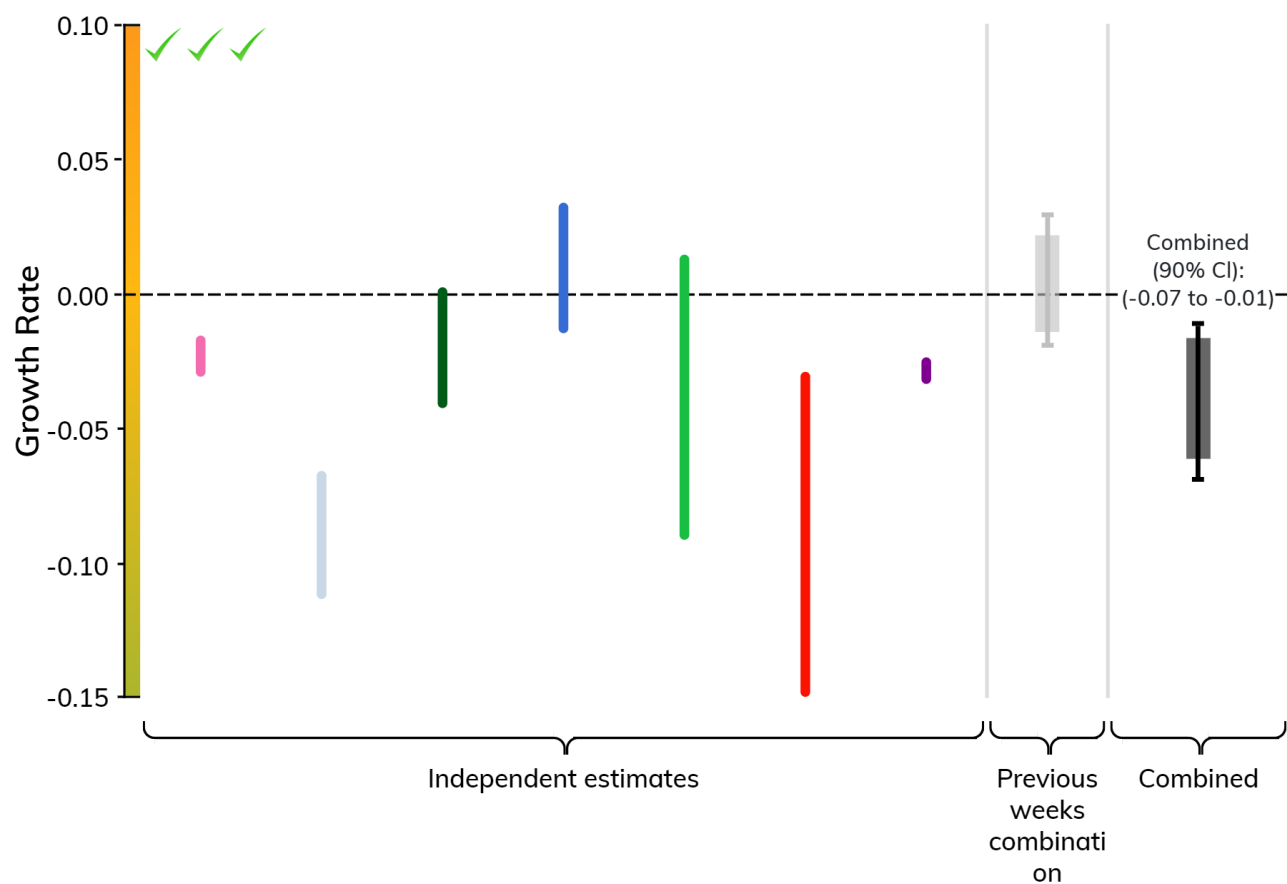


Figure 2c. Estimates of the growth rate in the 4 UK nations, including 90% credible intervals as at 12 April 2022

Bars represent different independent estimates. The grey shaded areas represent the combined numerical range and the black bars are the combined range after rounding outwards to the nearest per cent.

Scotland

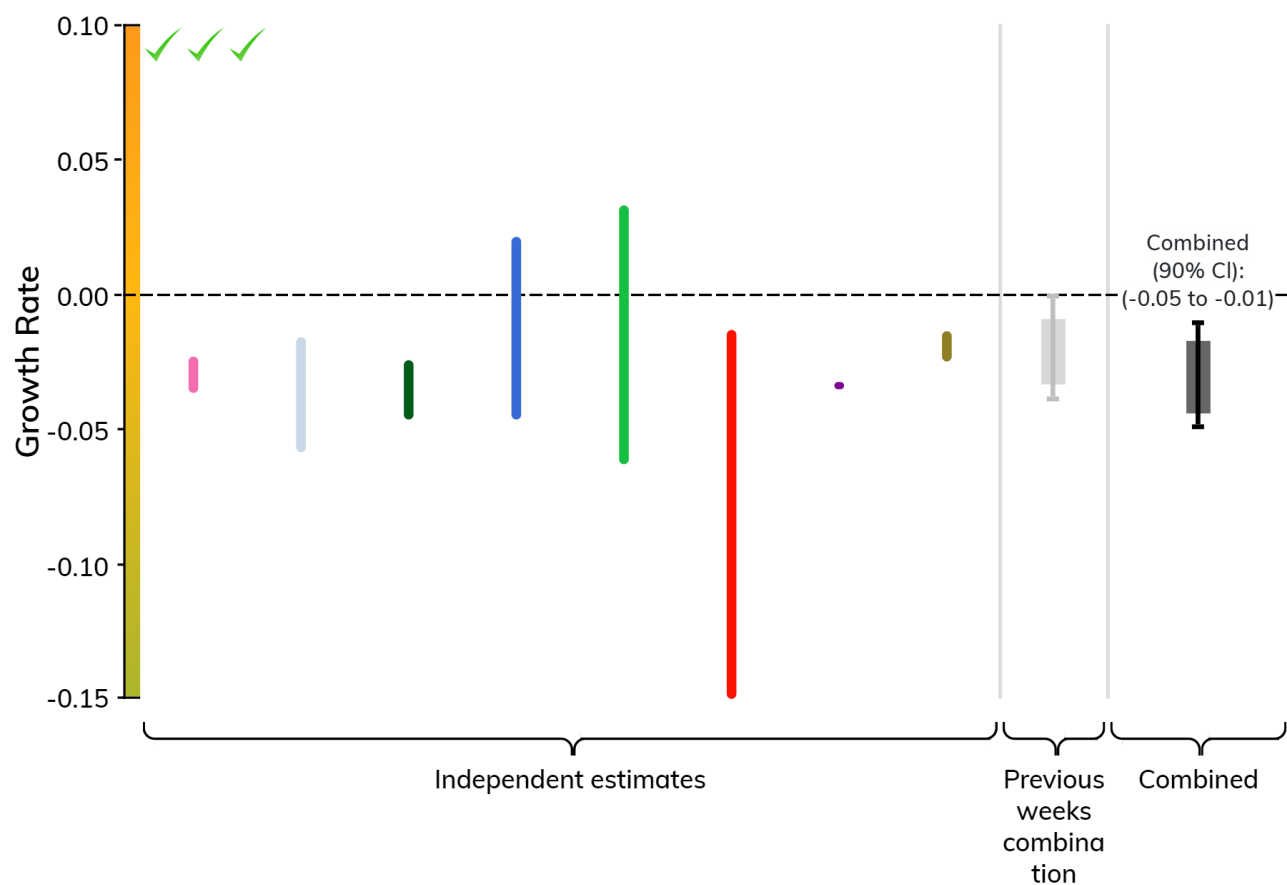


Figure 2d. Estimates of the growth rate in the 4 UK nations, including 90% credible intervals as at 12 April 2022

Bars represent different independent estimates. The grey shaded areas represent the combined numerical range and the black bars are the combined range after rounding outwards to the nearest per cent.

Northern Ireland

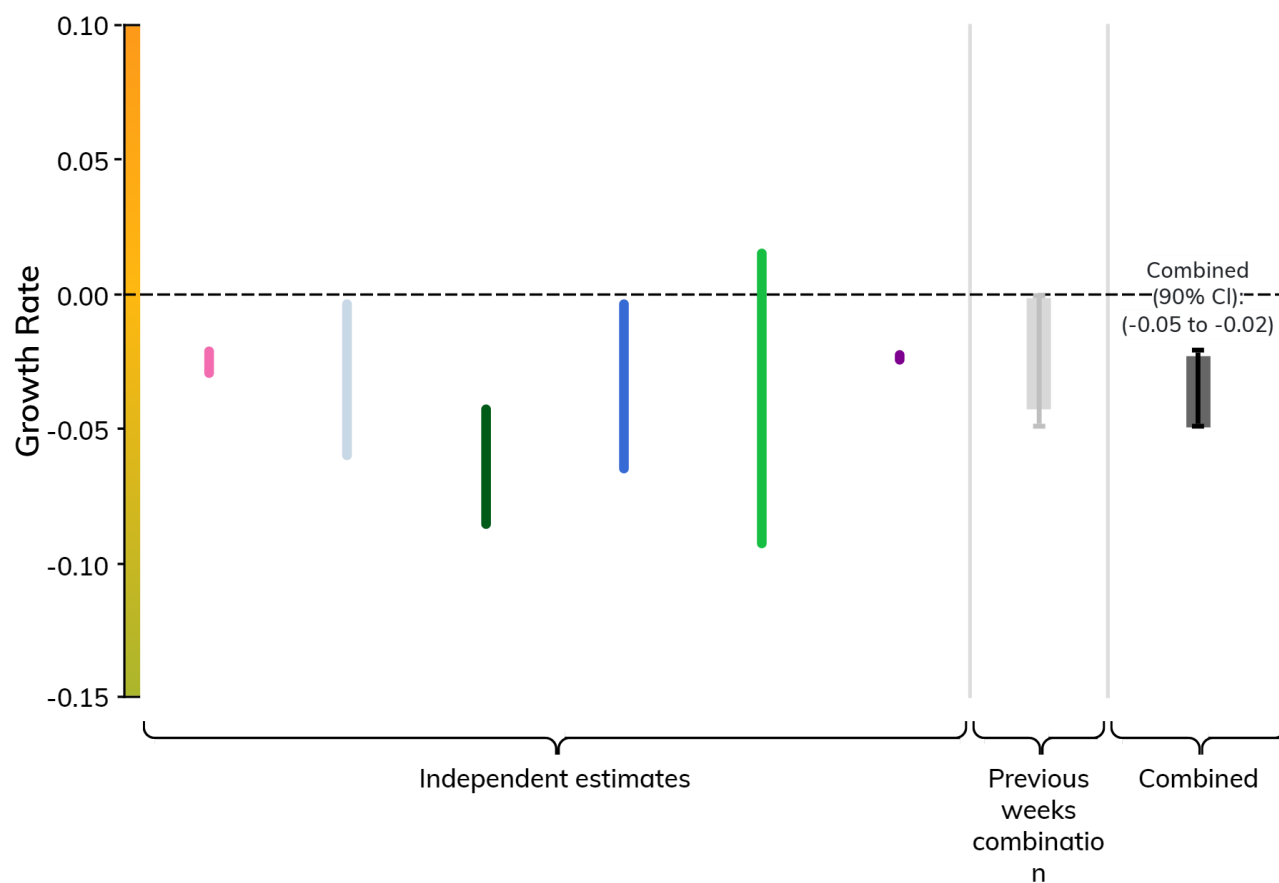


Figure 3a. Estimates of R in the NHS England regions, including 90% credible intervals as at 12 April 2022

Bars represent different independent estimates. The grey shaded areas represent the combined numerical range and the black bars are the combined range after rounding outwards to one decimal place.

London

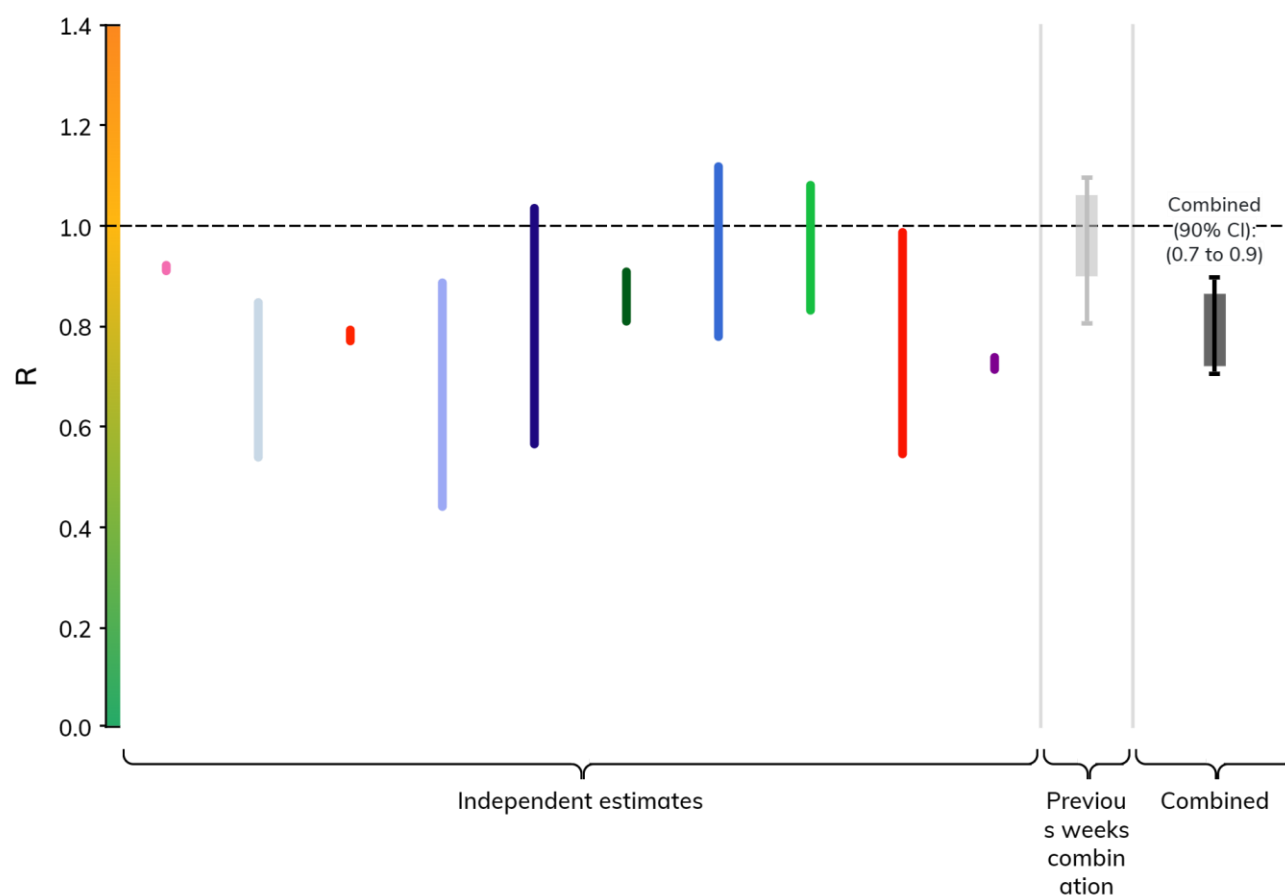


Figure 3b. Estimates of R in the NHS England regions, including 90% credible intervals as at 12 April 2022

Bars represent different independent estimates. The grey shaded areas represent the combined numerical range and the black bars are the combined range after rounding outwards to one decimal place.

East of England

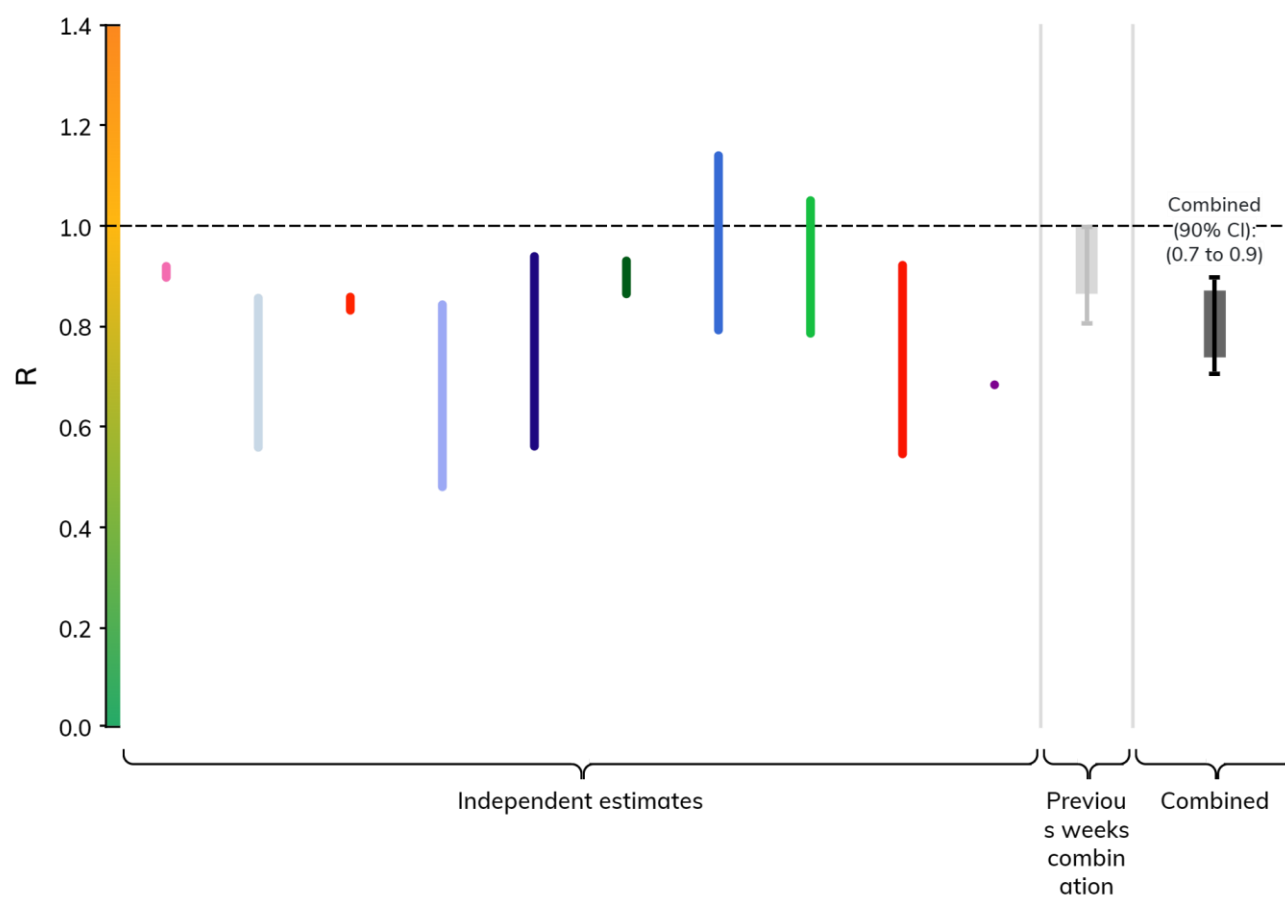


Figure 3c. Estimates of R in the NHS England regions, including 90% credible intervals as at 12 April 2022

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Midlands

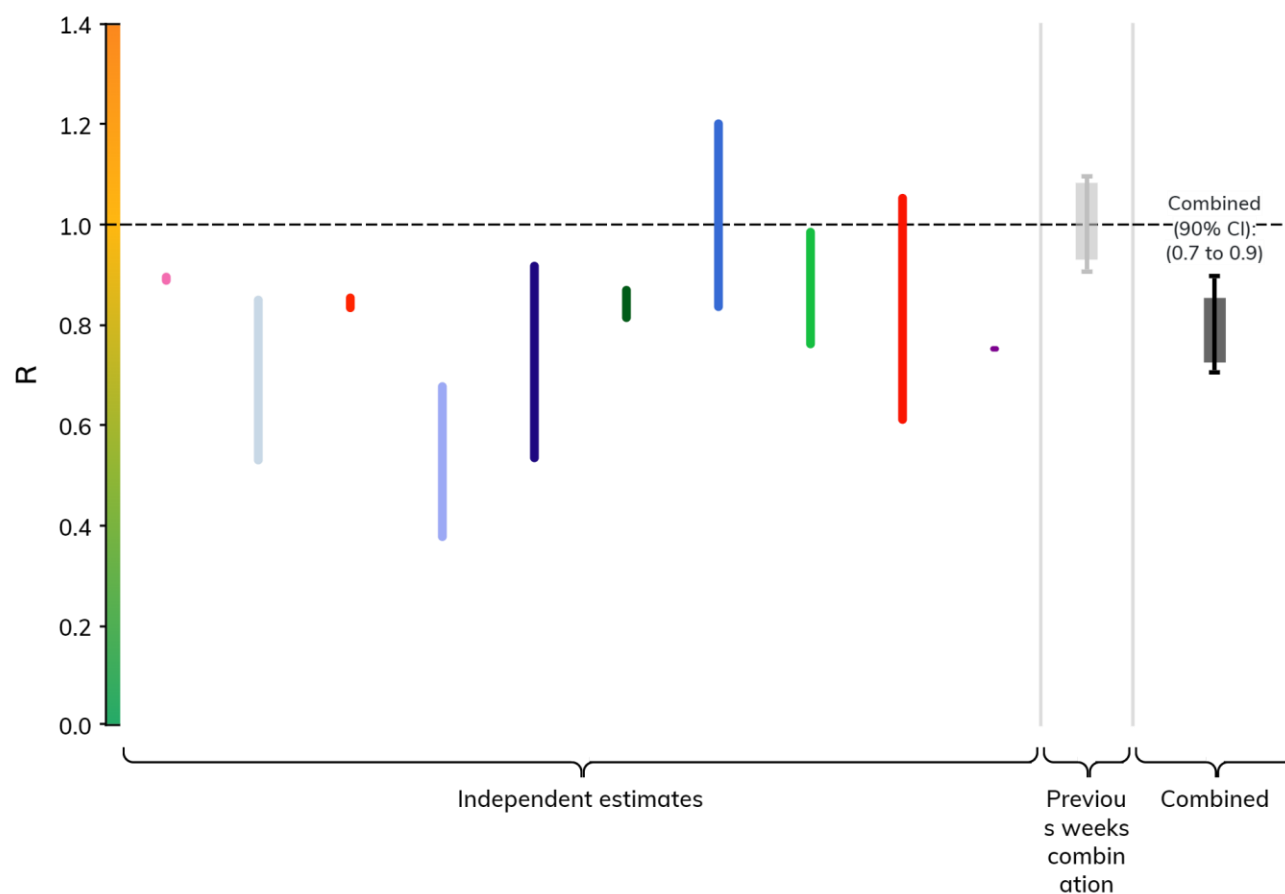


Figure 3d. Estimates of R in the NHS England regions, including 90% credible intervals as at 12 April 2022

Bars represent different independent estimates. The grey shaded areas represent the combined numerical range and the black bars are the combined range after rounding outwards to one decimal place.

North East and Yorkshire

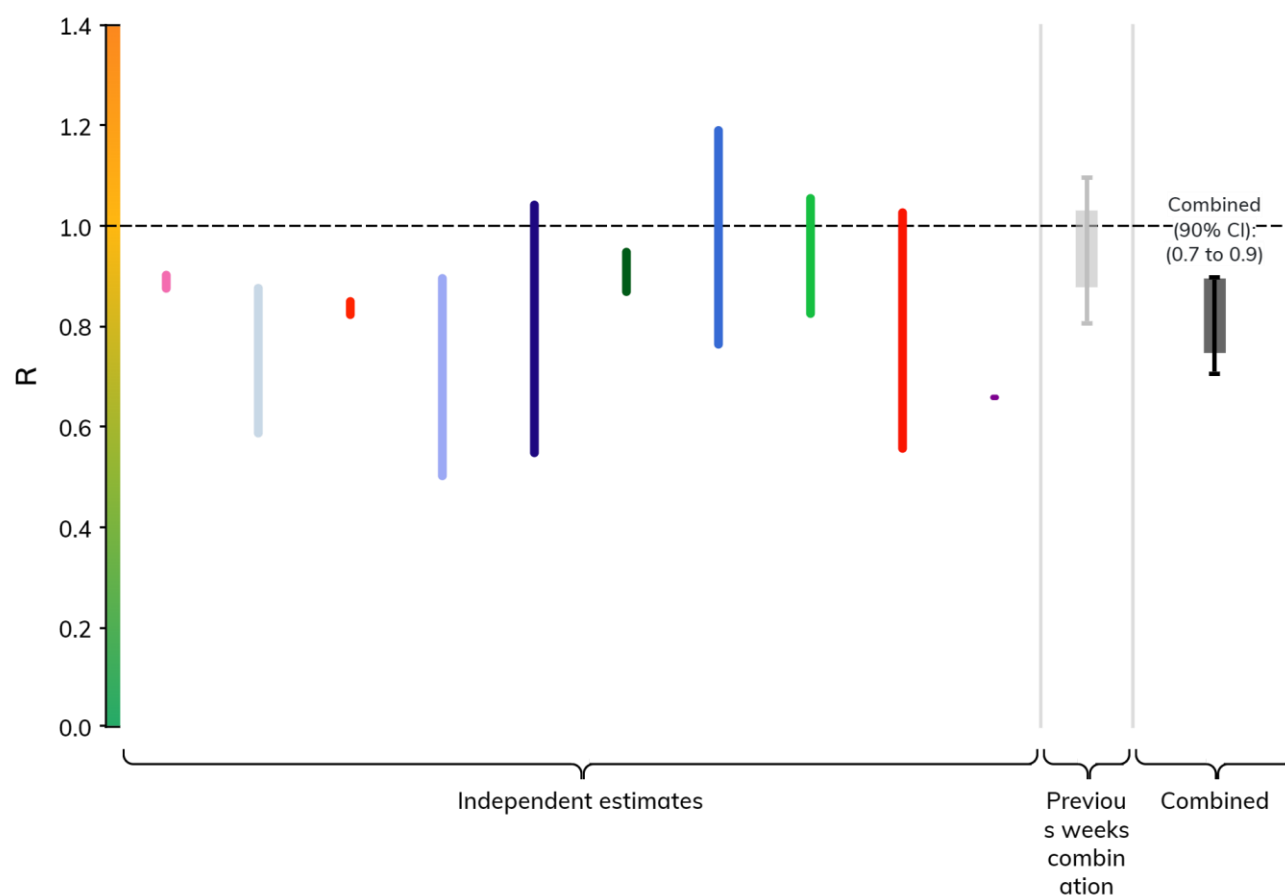


Figure 3e. Estimates of R in the NHS England regions, including 90% credible intervals as at 12 April 2022

Bars represent different independent estimates. The grey shaded areas represent the combined numerical range and the black bars are the combined range after rounding outwards to one decimal place.

North West

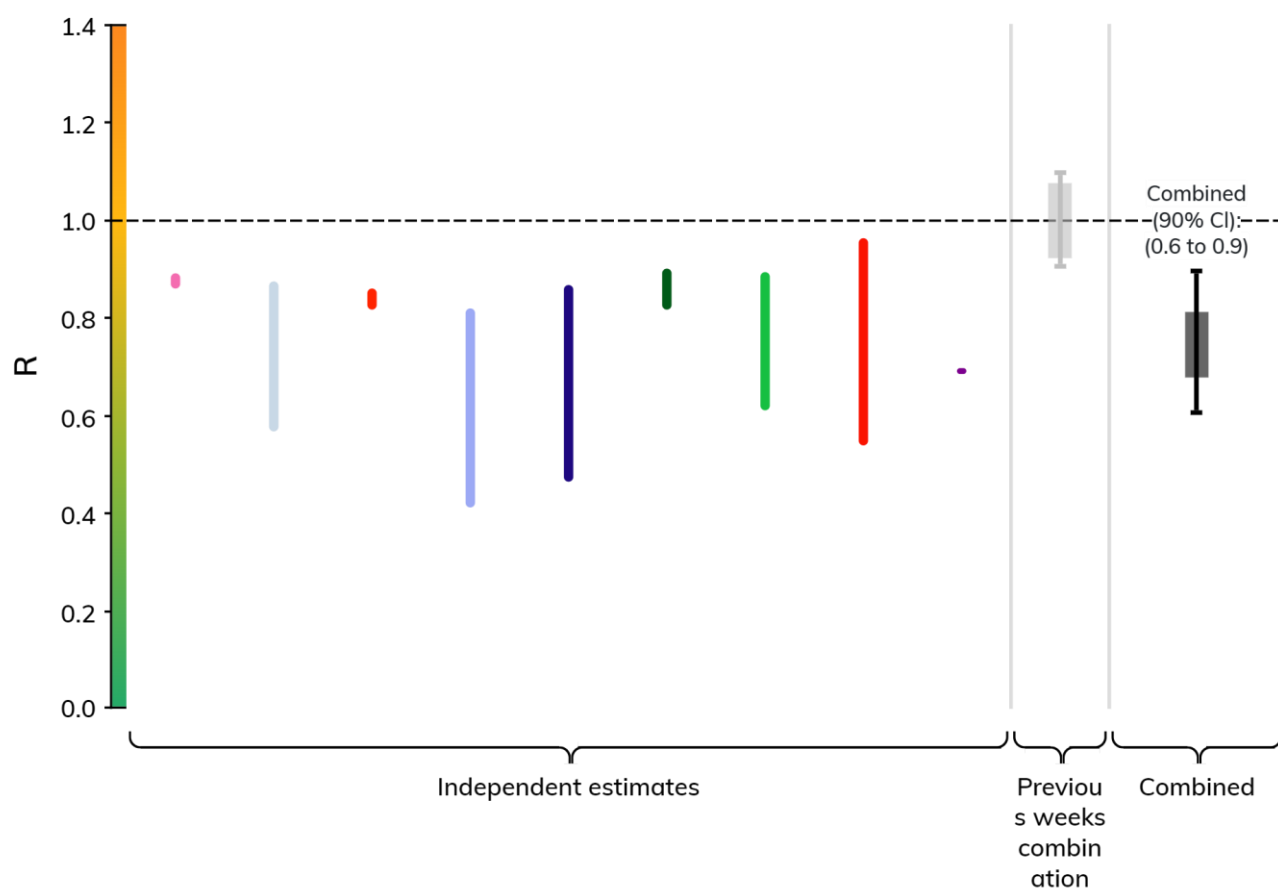


Figure 3f. Estimates of R in the NHS England regions, including 90% credible intervals as at 12 April 2022

Bars represent different independent estimates. The grey shaded areas represent the combined numerical range and the black bars are the combined range after rounding outwards to one decimal place.

South East

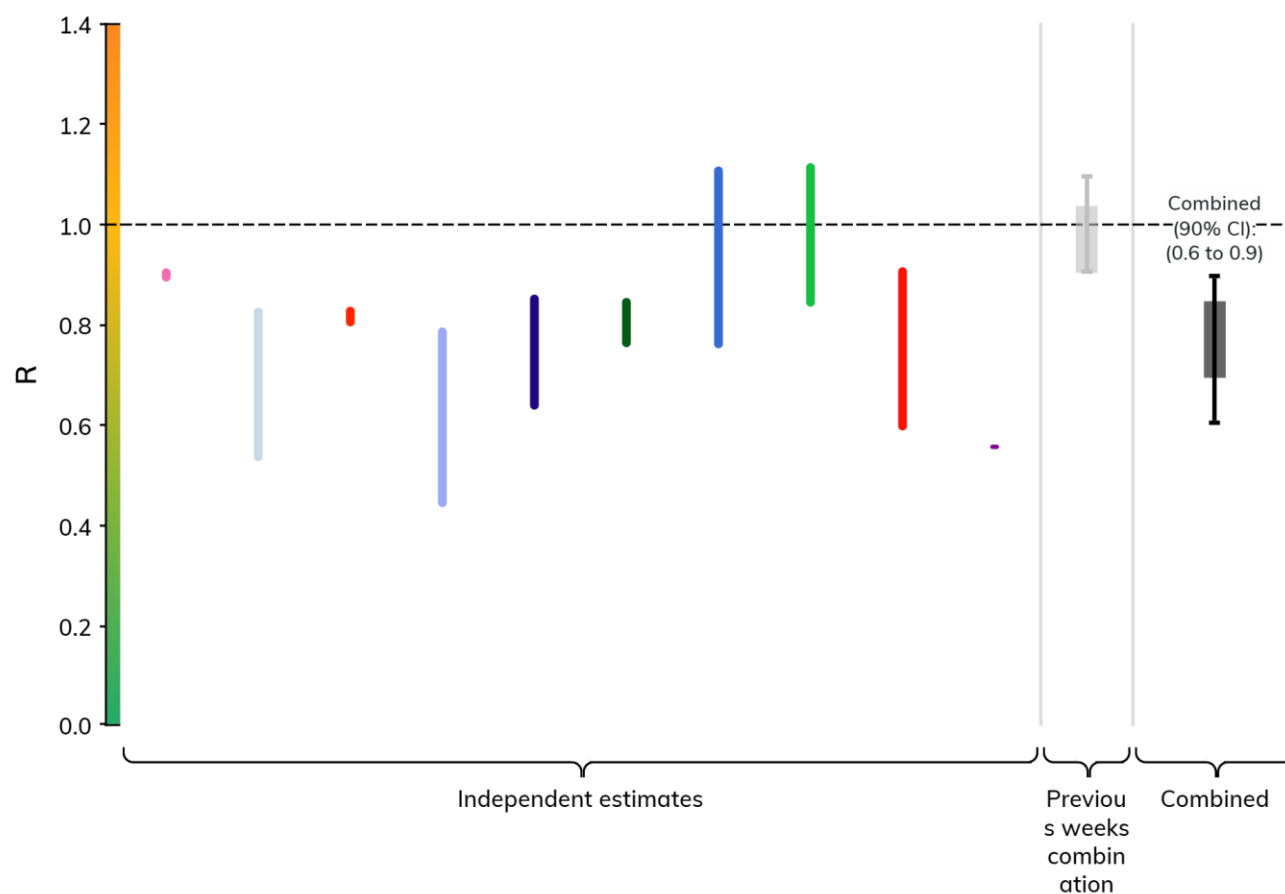
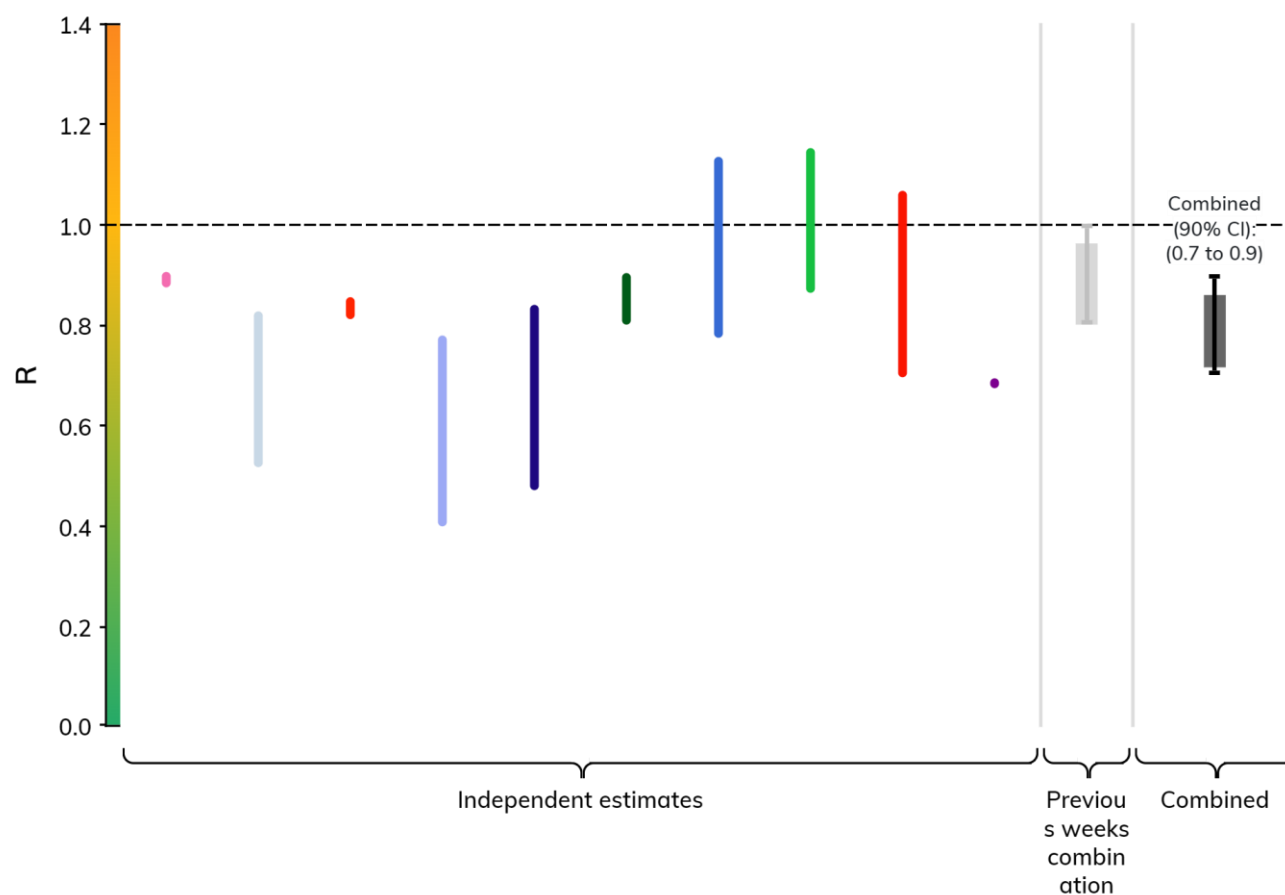


Figure 3g. Estimates of R in the NHS England regions, including 90% credible intervals as at 12 April 2022

Bars represent different independent estimates. The grey shaded areas represent the combined numerical range and the black bars are the combined range after rounding outwards to one decimal place.

South West



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